

Assignment 2
Subject: Operating System

Question No	Question																		
1	Explanation of process in operating system. Also, explain process cycle with suitable diagram.																		
2	Explain thread in operating system. Differentiate between Thread and Process?																		
3	Differentiate User level Threading and Kernel level Threading?																		
4	Explain process scheduling and various process scheduling queues.																		
5	Apply First Come First Serve (FCFS) process scheduling algorithm. Assume suitable data.																		
6	Apply Shortest Job First (SJF) process scheduling algorithm. Assume suitable data.																		
7	Apply Round Robin (RR) process scheduling algorithm. Assume suitable data.																		
8	Explain the working of Scheduler. Explain Various types of Schedulers.																		
9	Explain Inter-process communication with suitable example.																		
10	Describe the various operations that can be performed on process and thread.																		
11	Explain multiprocessor scheduling with suitable example. Design your own CPU scheduling algorithm which works in multiprocessor environment.																		
12	Describe various fields of process control block with a neat sketch.																		
13	List all the criteria for evaluating cpu scheduling algorithm. Which one is best among FCFS, SJF and RR considering each parameter?																		
14	Explain Context Switching with example																		
15	Differentiate between Process and Program																		
16	Draw Gantt charts to illustrate the execution of the processes using the following scheduling algorithm: <table><tr><td><u>Process</u></td><td><u>Burst Time</u></td><td><u>Priority</u></td></tr><tr><td>P_1</td><td>2</td><td>2</td></tr><tr><td>P_2</td><td>1</td><td>1</td></tr><tr><td>P_3</td><td>8</td><td>4</td></tr><tr><td>P_4</td><td>4</td><td>2</td></tr><tr><td>P_5</td><td>5</td><td>3</td></tr></table> (1) FCFS, (2) nonpreemptive SJF, (3) preemptive SJF, (4) nonpreemptive priority, (5) preemptive priority, and (6) RR with time quantum=2	<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>	P_1	2	2	P_2	1	1	P_3	8	4	P_4	4	2	P_5	5	3
<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>																	
P_1	2	2																	
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17	A process in the lower priority queue can suffer from starvation due to some short processes taking all the CPU time. Suggest a possible solution.																		
18	Consider a system which has a CPU bound process, which requires the burst time of 40 seconds. The multilevel Feed Back Queue scheduling algorithm is used and the queue time quantum '2' seconds and in each level it is incremented by '5' seconds. Then how many times the process will be interrupted and on which queue the process will terminate the execution?																		
19	Multilevel Feedback Queues: 3 jobs, of length 30, 20, and 10 seconds, the 10 sec job has 1 sec of I/O every other sec, initial time slice 2 sec, context switch time of 0 sec, 2 queues.																		
20	Define process and Explain process states in details with diagram																		
21	Explain process states and process control block in details																		
22	Explain the process state transition diagram used in multiprogramming environment.																		

23	Describe the fields in a process control block (PCB).What is switching overhead?										
24	What is thread? Explain classical thread model										
25	Distinguish between preemptive and non-preemptive Scheduling										
26	List the main difference and similarities between threads and process.										
27	What are various criteria for a good process scheduling algorithm? Explain any two preemptive scheduling algorithms in brief.										
28	Explain the effect of increasing the time quantum to an arbitrary large Number and decreasing the time quantum to an arbitrary small number for round robin scheduling algorithm with suitable example?										
29	Consider following processes with length of CPU burst time in milliseconds <table> <tr> <td>Process</td><td>Burst time</td></tr> <tr> <td>P1</td><td>5</td></tr> <tr> <td>P2</td><td>10</td></tr> <tr> <td>P3</td><td>2</td></tr> <tr> <td>P4</td><td>1</td></tr> </table> All process arrived in order p1, p2, p3, p4 all time zero a) Draw Gantt charts illustrating execution of these processes for SJF and round robin (quantum=1) b) Calculate waiting time for each process for each scheduling algorithm c) Calculate average waiting time for each scheduling algorithm	Process	Burst time	P1	5	P2	10	P3	2	P4	1
Process	Burst time										
P1	5										
P2	10										
P3	2										
P4	1										
30	Differentiate between process and threads										
31	Explain about different multithreading models										
32	Define Thread. Define Time slice. What does PCB contain?										
33	Write the difference between user level thread and kernel level thread? Under what circumstances user level threads are better than the kernel level threads?										
34	What is meant by context switch?										
35	What are the use of job queues, ready queues and device queues?										
36	Compare and contrast Single-threaded and Multi-threaded process.										
37	Priority inversion is a condition that occurs in real time systems – Analyzing on this statement.										
38	Distinguish between CPU bounded, I/O bounded processes.										
39	What resources are required to Creating threads?										
40	What is the meaning of the term busy waiting? Define the term ‘Dispatch Latency’.										